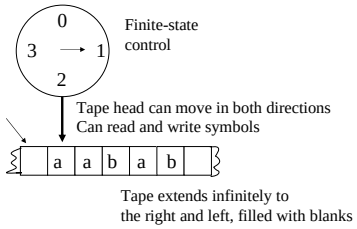
CS 3813: Introduction to Formal Languages and Automata

Turing machines (9.1)

Turing machine

(Alan Turing 1936)



All of the automata we study in this class have a finite number of internal states.

They differ in the "auxiliary memory" they have and how it is organized.

DFA, NFA	finite memory
DPDA, PDA	infinite stack memory
Turing machine	infinite tape memory (tape can be read forwards and backwards without erasing)

Turing machine: formal definition

- Q is finite set of internal states
- q_0 is initial state
- $F \subseteq Q$ is set of final states
- Σ is input alphabet
- Γ is tape alphabet (includes all symbols in input alphabet, special blank symbol, and maybe others)
- $\delta : Q \times \Gamma \rightarrow Q \times \Gamma \times \{L, R\}$ is transition function

Transitions

- A transition of the Turing machine has the form
  $\delta(q, a) = (q', b, L)$,
- where:
  - q is the current state,
  - q' is the state the TM makes a transition to,
  - a is the current symbol read from the tape,
  - b is the symbol the TM writes on the tape, which could also be the blank symbol, ♦
  - L means move the tape head one cell to the left (and R means move the tape head one cell to the left)
- For now, we assume the Turing machine is deterministic (a DTM), which means that δ defines at most one move for each configuration

Input string

- As before, the input string is on the initial tape, and the read/write head is initially positioned over the leftmost symbol of the input string
- The rest of the tape is filled with blanks
- A TM does not have to read the string in one direction and stop when it finishes reading
- It can move forward and backwards over the input string, as many times as it wants
- It can also overwrite the input string, as well as write anything else it wants on the tape

Halting

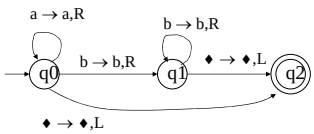
- When does a TM halt?
  - when it reaches a configuration for which there is no transition defined
  - when it enters a final state (we will assume no transitions are defined for a final state)
- It is important to note that a TM does not necessarily halt. It can enter an infinite loop.

Acceptance

- We can define how a TM accepts a string in different ways.
- We will say that it accepts a string if it halts in final state. It rejects the string if it halts in some other state, or does not halt at all.

Example

The following diagram represents a TM that accepts the language denoted by the regular expression a*b*.



How do you write the transitions for this?

Exercises

- Recall that the following languages are not context-free.
  - $\{a^n b^n c^n | n \geq 0\}$
  - $\{ww \mid w \in \{a,b\}^*\}$
- In English, describe how a Turing machine could accept each of these languages
- Try to draw the transition diagrams for the Turing machines

Model of computation

- A TM is an abstract model of a computer that lets us define in a precise, mathematical way what we mean by computation.
- As before, we use the concept of a "configuration" to define what we mean by computation.

Configuration

- A configuration of a TM includes everything we need to know to continue a computation
  - current state
  - symbol currently under the tape head
  - string on tape to left of the head (up to leftmost non-blank symbol)
  - string on tape to right of the head (up to rightmost non-blank symbol)
- Example
  - (q, a, abb, bbb)

Computation

- A computation is a sequence of configurations such that each configuration is obtained from the previous configuration by some transition of the TM

Computing functions

- We use automata to recognize languages *and compute functions* (although we haven't talked about computing functions until now).
- In fact, computing functions allows language recognition, since every language can be represented by a *characteristic function*
  - if the input string is in the language, output 1
  - otherwise output 0

String and numeric functions

- A TM can compute functions on strings. Given an input string, the output of the function is the string left on the tape after the TM halts.
- A TM can also compute numeric functions. Using a binary alphabet for strings, the input string can represent a binary number and the output string can represent a binary number. (A unary alphabet can also be used, as in the book.)
- A TM can compute a function with multiple arguments. Each argument is separated by a special character, delimiter, on the tape.

Transducers

- We can create TMs that compute arithmetic functions such as addition, subtraction, multiplication, etc.
- A TM that computes other things besides language acceptance is called a transducer. A transducer computers some output (which could be anything) for some input.

Exercise

Construct (or describe how to construct) a TM that computes the numeric function $f(k) = k + 1$.